What the NIST factoring numbers mean

R08 finished at 2026-10-25 07:52:38 UTC. It collected more raw data and found no new factor. No factoring process is currently running. The next experiment is defined separately for later execution.

The answer so far

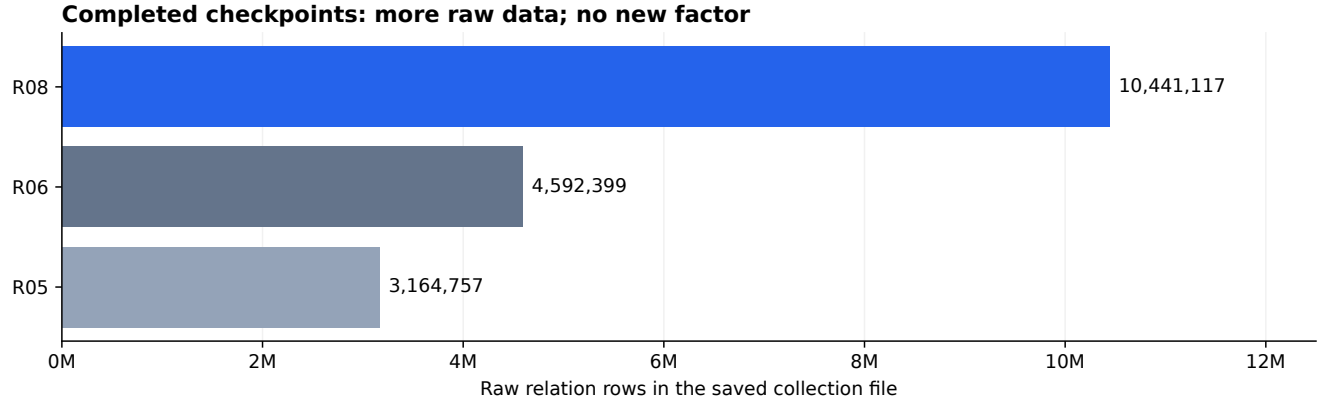
Number	What is counted	What it establishes
4 skipped curves	P-224 candidates 777, 1485 and 5736; P-256 candidate 876	The original curve population, not the number of factoring targets
9 target integers	Distinct integers behind the imported DiSSECT factorization timeouts, including published control curves	8 have complete prime-certified factorizations
14 timeout fields	Report fields that sometimes request the same integer	13 are resolved; this is a different denominator from 9
444 bits	Size of the remaining composite cofactor	One unresolved integer; not a security level or a runtime prediction

The outstanding target is skipped_256_876-twist2: a 512-bit degree-2 extension-twist order associated with P-256 candidate 876. It is not the 256-bit base-field curve order or its base-field quadratic-twist order. Its certified partition remains 1657 × 45294121 × 5333949541 × C, where C is the exact 444-bit composite in the result. The factorization scope explains the imported timeout fields. Resolving these arithmetic fields does not establish why a seed was historically skipped.

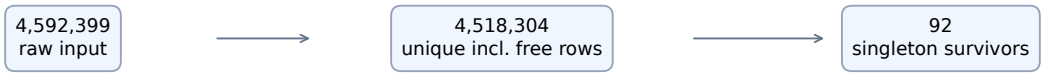
What the relation counts mean

A raw NFS relation is a recorded arithmetic relation produced by sieving. It is input to later processing. Rows can repeat, fail an integrity check, or fail to combine into the dependencies needed for linear algebra. More raw rows are more collected data; they are not discovered prime factors.

Collection checkpoint	Raw lines in its main relation file
R05	3,164,757
R06	4,592,399
R08	10,441,117



R07 filtered the older R06 file — not the R08 file



R08 filtering: NOT RUN • raw counts do not establish matrix rank or percentage complete

For R08, 4,592,399 inherited + 5,848,718 newly collected = 10,441,117 raw lines. The native log counts only the new allocation; wc -l counts the complete saved file. Both are correct and have different scopes. The inherited prefix has the same SHA-256 as R06's original file. The final file is 919,799,315 bytes, SHA-256 62106d7ae9e828ff433d019ba471cf743552b608421e889cd83f6af0296f20b0.

The native request was 41,407,601 additional relations, corresponding to a configured total of 46,000,000 after the inherited file. That is the tool's collection request. It is not a verified matrix threshold or a percentage-complete denominator. The native ETA is not admitted as a factoring completion estimate.

Why 4.59 million rows previously became only 92

R07 filtered an independent copy of R06, before R08 collected its extra data. Its native log records:

Stage	Recorded count	Meaning
Raw input	4,592,399	Original collected rows
Rejected rows	1	A row with composite factors was discarded; filtering continued
Free relations added	78,556	Extra relations generated by the filter, not extra sieving
Duplicates removed	152,650	Repeated relations removed
Unique relations	4,518,304	4,592,399 − 1 + 78,556 − 152,650
Tracked large ideals before singleton removal	10,585,421	Several ideals may occur in one relation; these are not distinct integer factors
Relations left after singleton removal	92	Four pruning passes; no matrix prerequisite was produced

An ideal appearing in only one remaining relation cannot cancel in a dependency. Removing its relation can expose another singleton, so pruning cascades. Here the filter tracked ideals above its cutoff of 100,000 and ended with 0 tracked large ideals. That does not mean the entire integer factorization was solved. The native controller requested more relations; rank and a usable matrix were not established.

The 92 count applies to the older 4.59-million-row input. The newer 10.44-million-row checkpoint has not been filtered. Applying the old survival fraction to the new file would be an unsupported extrapolation.

Execution, verification and provenance

One R08 allocation was planned, attempted and completed; none was excluded. The frozen R08 protocol and actual command bind the input, seed 20261022, two workers, native 7200-second budget, outer 7260 seconds plus 30-second kill grace, 4 GiB virtual memory per process and 1 GiB per output file. The native sieve began at 05:52:08 UTC and reported its limit after 7229.9807 seconds while outstanding worker work stopped. The supervisor returned exit 0; the outer process timeout did not fire. This is a normal budget stop, not factoring completion.

This was an L0 shared local host. UTC timestamps and native elapsed output are provenance, not comparative timing evidence. Runtime source was blob-checked against commit 5cb553810e5bbfbd8691ec614ffcaacefc625757. Patched YAFU SHA-256 is bc7680adbdd7450a68120e73ef8aa88d0336c8ffa141343f50966c595ca6bae9; the earlier binary-bound timeout controls remain applicable.

The post-run native verifier and a fresh replay on October 26 confirm the whole target, exact product and retained prime certificates. A partial partition is not a complete prime factorization. Existing Lean partition identities remain unchanged; no new algebraic identity is claimed. The October 26 FactorDB response still lists C with no proper split.

Small logs, receipts, certificates, checksums and the result are committed. The bulk manifest identifies the full local archive; its remote publication is blocked by the existing artifact-access problem. No remote object or successful upload is asserted. A second-runner review of this completion record has not been performed.

The experiment submitted for later execution

XP-20261026-nist-p256-checkpoint-filter uses the repository's backlog, frozen experiment.protocol/v1 record and draft crypto.experiment/v1 tracker. It first replays R06's known filtering counts, then runs the same filter on a separate R08 copy. Neither source checkpoint is modified. Its 600-second native and 660-second outer limits apply equally to both arms. It stops before new sieving or linear algebra.

The result will distinguish a completed native request for more relations, new matrix prerequisites requiring review, and an incomplete execution. It will preserve every status and return a unified experiment.export/v1; it will not translate raw counts into a security or historical-selection claim. The card is blocked until the input artifacts and execution setup are available. No new experiment has been executed or queued by this PR.

Sources and graph scope

Reviewed primary implementation sources: YAFU c7d90283972820f99b3b4920bfc1eff05c1f6342, factor/nfs/nfs_postproc.c (native filtering), factor/nfs/nfs.c (controller response), and gnfs/filter/{filter,duplicate,singleton}.c. The upstream source and retained R07 logs explain why collection, filtering and matrix readiness are separate stages. No new factoring method is proposed. The current backlog protocol, queue design, and canonical experiment protocol define the prospective submission.

The new graph plots recorded counts, one completed checkpoint per bar, without a timing interval or performance ratio. Prior R05/R06/R07/R08 interim graphs stay as dated history. IC leaderboards and curve/isogeny graphs are unchanged: no cost comparison, new curve relation, security finding or explanation of the skipped seeds was established.